Fleet Rebalancing with Multi-Agent Reinforcement Learning: A Study on Synthetic Demand (Bandra) and Real Demand (NYC16 Baselines)

Alaa Daoud

LAMIH, CNRS, INSA Hauts-de-France, UPHF

Avani Dewangan, Md Athar Sharif, Prashil Thul
IIIT Naya Raipur, B.Tech CSE 2024–2028

October 29, 2026

Abstract

We train IPPO and MAPPO zone-controller policies on a synthetic 25-zone network (Bandra, 30 drivers, $\beta = 0.1$, $\gamma = 5.0$) and evaluate them against greedy and LP baselines under a standardized demand-shift taxonomy (S0–S7). On the five-seed final-validation protocol, RL matches greedy on service rate while spending roughly 8× greedy's deadhead, and the LP baseline dominates RL on the service-rate/deadhead tradeoff (higher service rate at $\sim 0.58\times$ RL deadhead). Because Bandra and NYC16 have different observation spaces, the NYC16 results are baselines only and are not directly comparable to the Bandra RL results. All reported numbers are reproducible from artifacts committed in `results/` and `runs/`.

1 Introduction

Problem. Fleet rebalancing is the decision of where to move idle drivers so that they meet demand before riders cancel. The study asks two questions: (RQ1) does learned relocation beat a tuned LP baseline on the service-rate / empty-distance tradeoff, and (RQ2) does the algorithm (IPPO vs. MAPPO) matter once the reward is changed?

Environment. Two networks. **Bandra**: 25 synthetic zones on a 5×5 grid, 30 drivers, 30-minute episodes, 200 orders, 5-minute patience. **NYC16**: the 16 busiest taxi zones from real FHVHV data, 26 drivers, 130 orders/hour, 5-minute patience. The two are not interchangeable: Bandra checkpoints cannot load on NYC16 because the observation space differs. All RL training and all RL results in this paper are on Bandra; NYC16 appears only as a baseline comparison.

Baselines. GreedyZoneDispatcher (nearest-idle dispatch, zero relocation) and ZoneLPRebalancer, a rolling-horizon transportation LP adapting MIVR [1] with an EMA demand forecast and a 5-minute lookahead; parameters in Section 2.

Method. IPPO and MAPPO with a destination-conditioned relocation head, trained with PPO in a PettingZoo AEC environment [2] and evaluated deterministically with argmax actions. The team reward is $R = \gamma \cdot \text{served} - \beta \cdot \text{empty km} - \alpha \cdot \text{wait} - \text{cancel}$, with α=0.2, β=0.06, γ=1.0 on NYC16 and α=0.1, β=0.1, γ=5.0 on Bandra. A fixed 65% share of the served credit is attributed to the relocating zone; the Bandra configuration pays this on top of γ rather than carving it out.

Headline. On the five-seed final-validation protocol, RL matches greedy on service rate while spending roughly 8× greedy's deadhead, and the LP baseline dominates RL on the tradeoff (higher service rate at $\sim 0.58\times$ RL deadhead). The deadhead multiplier is protocol-dependent; Section 4 quantifies the spread.

2 Environment and Baselines

2.1 Bandra

25 synthetic zones on a 5×5 grid (Bandra West, Mumbai). 30 drivers, 30-minute episodes ($dt = 0.5$ min, 60 turns), 200 orders, 5-minute order timeout, 30 km/h speed. Reward constants: $\alpha = 0.1$, $\beta = 0.1$, $\gamma = 5.0$, cancel $= 0.5$; the relocation bonus is paid on top of γ rather than carved out of it. 1,500 zone-turns per episode and 9,000 batch-1 forwards per training iteration.

2.2 NYC16

The 16 busiest taxi zones from real FHVHV 2026-04 data (zone IDs 68, 79, 112, ...). Zone centroids from the Manhattan road network, 84 train / 12 val / 12 test windows. 26 drivers, 130 orders/hour, 5-minute patience. Reward constants $\beta = 0.06$, $\alpha = 0.2$, $\gamma = 1.0$, relocator share 0.65.

2.3 Baselines

GreedyZoneDispatcher: nearest-idle dispatch, zero relocation. **ZoneLPRebalancer**: rolling-horizon transportation LP with EMA demand forecast (smoothing $\alpha = 0.3$, 5-minute lookahead, deficit penalty 1.0), tuned on eight TRAIN windows on NYC16.

2.4 Incompatibility

Bandra checkpoints use an 872-dimensional observation vector (25 synthetic zones, $\alpha = 0.1$, $\beta = 0.1$, $\gamma = 5.0$). NYC16 uses a different observation dimension and $\beta = 0.06$. They cannot be loaded interchangeably, so the two tables measure different environments.

Implementation. All 30 training cells (IPPO and MAPPO × 40/80/200 iterations × 5 seeds) ran on CPU. Each update splits six rollout episodes across four independent environment streams in a 2/2/1/1 pattern, so runs are not bitwise identical to a serial implementation. All reported evaluations use deterministic argmax actions.

3 Training Results

3.1 Setup

IPPO and MAPPO × 40/80/200 iterations × 5 seeds, all on Bandra (25 zones, 30 drivers), held-out validation, deterministic argmax evaluation. Final-iteration numbers are reported (not best-

selected). All standard deviations are sample std (ddof = 1) across the 5 seeds.

3.2 Final-iteration validation results

Arm	SR	Empty km	Wait
IPPO-40	33.2% ± 1.3	185.8	2.3 min
IPPO-80	34.1% ± 0.6	182.2	2.3 min
IPPO-200	34.1% ± 1.0	170.6	2.2 min
MAPPO-40	34.0% ± 1.0	168.5	2.1 min
MAPPO-80	33.7% ± 1.3	179.3	2.2 min
MAPPO-200	35.1% ± 0.3	167.8	2.2 min

3.3 Convergence

200-iteration arms are flat over the last 10 checkpoints: IPPO-200 mean 34.1% (range 32.9–35.5 across checkpoints 110–200); MAPPO-200 mean 35.2% (range 33.4–36.6). The 200- and 80-iteration arms are within 1 pp, confirming a plateau. IPPO ≈ MAPPO at $\beta = 0.1$: no divergence, so RQ2 is answered negatively.

4 Argmax vs. Sampled Evaluation

4.1 Protocol

IPPO-200-s0 and MAPPO-200-s0, 50 episodes each, base seed 2000, deterministic versus stochastic policies on identical demand windows. Artifact: `results/bandra_argmax_pilot/`.

4.2 Results

Algo	Policy	SR (± std)	Empty km (± std)
IPPO	argmax	35.9% ± 3.4	95.1 ± 16.6
IPPO	sampled	36.7% ± 3.2	153.0 ± 15.5
MAPPO	argmax	36.5% ± 2.6	150.2 ± 15.3
MAPPO	sampled	36.8% ± 2.4	160.8 ± 13.8

ΔSR: IPPO −0.8 pp, MAPPO −0.4 pp. Argmax cuts empty km by 57.9 km for IPPO and 10.6 km for MAPPO per episode at no cost in service rate.

4.3 Decision

IPPO's SR difference is within 1 pp, so we select argmax for its 58 km deadhead reduction. MAPPO also prefers argmax, but only by 10.6 km. All evaluation uses argmax going forward.

4.4 Deadhead depends on the evaluation protocol

Deadhead is not a property of a checkpoint alone; it depends on the demand windows and the fleet size at evaluation time. The numbers below all concern IPPO-200:

- 95.1 km: pilot evaluation, 50 episodes over seeds 2000–2049, 30 drivers.

- 146.1 km: seed-s0 `final_validation.json`, 12 episodes at eval seed 9000, 30 drivers.
- 170.6 km: five-seed mean over seeds s0–s4, each 12 episodes at eval seed 9000 (sample std 14.4 km); this is the protocol used for the training and baseline tables.
- 83.4 km: fleet sweep at 29 drivers, 10 episodes, base seed 3000.
- 136.5 km: shift-eval S0 mean (5 seeds × 5 episodes, seeds 1000–1004), the left-panel S0 point in Figure 1.

The pilot's 95.1 km is a single checkpoint on seeds 2000–2049 and is not paired with LP's 98.4 km, which is measured on seeds 9000–9011. The 8× headline uses 170.6 km against greedy's 20.7 km on the five-seed final-validation protocol. On the pilot windows the same checkpoint is 95.1 km, close to LP's 98.4 km, so the multiplier is evaluation-dependent: the defensible claim is that RL spends several-fold more deadhead, with 8× specific to the five-seed protocol.

5 Shift Evaluation

5.1 Taxonomy

Eleven standardized shift scenarios (S0–S7), evaluated on Bandra synthetic demand and on NYC16 with baselines. Magnitude shifts (S2a–S2e) scale `num_orders` by `limit/130` on NYC16 so severity is a multiple of the in-distribution rate; on Bandra the base is 200 orders per episode and shifts use a proportional factor (e.g. S2a = 0.5 × 200 = 100 orders). S6 (NYC16 test split) is a no-op on Bandra and equals S0.

Label	Mechanism	Description
S0	—	Base (train)
S1	Permute	Spatial permutation (rot offset = $nz//2$)
S2a	×0.5	Magnitude ×0.5
S2b	×0.75	Magnitude ×0.75
S2d	×1.5	Magnitude ×1.5
S2e	×2.0	Magnitude ×2.0
S3	Temporal	Phase shift (val split)
S4	Concentrate	60% demand to cold zone z1
S5	Burst	20% in [20,30] min to z0
S6	Held-out	NYC16 test split; no-op on Bandra (equals S0)
S7	Drift	Hotspot A=zone12 → D=zone10

S2c (magnitude ×1.0) is intentionally not run: it is byte-identical to S0 and was dropped as compute waste. S3b is not run either.

5.2 Results (Bandra, 30 RL checkpoints)

Every arm is flat (∼35–38% SR) across S0, S1, S2a–S2e, S3, S5, S6, S7. On Bandra, S6 is a no-op: the "test split" switch only affects the NYC16 order source, so the Bandra S6 point is numerically identical to S0 in every run and is not evidence of held-out robustness. The only real degradation is S4 (60% of demand concentrated into one cold zone), where all arms collapse to ∼21% SR. IPPO-200 and MAPPO-200 are effectively identical (∼36% across all shifts); no divergence at $\beta = 0.1$. All shifts except S4 stay within 3 pp of S0.

5.3 Results (NYC16, baselines)

Method	S0 SR	S4 cold-zone	S6 held-out	Mean retention
Greedy (NYC16)	44.8%	22.6%	45.7%	91.5%
LP (NYC16)	69.7%	46.9%	69.4%	92.4%

LP dominates Greedy on SR across all shifts on NYC16; both degrade to ∼47% / ∼23% under S4.

5.4 Scope and comparability

Bandra RL checkpoints (25 zones, synthetic, 872-dimensional observation) cannot load on NYC16 (16 zones, real, different observation dimension), and no NYC16 RL results are reported in this paper. The two tables measure different environments and are not directly comparable. On Bandra, the Greedy vs LP baseline comparison uses paired demand windows (base seeds 9000–9011), so that comparison is paired. The RL arms are evaluated on a separate seed set, so the RL-versus-baseline comparison is not paired; Section 7 states the consequence.

6 Fleet-Size and Reward-Weight Sensitivity

6.1 Fleet-size sensitivity

Evaluation-time driver sweep on frozen IPPO-200-s0 and MAPPO-200-s0, 10 episodes each, base seed 3000. Directional signal only; policies are not retrained per fleet size. Artifact: `results/bandra_fleet_D93/`.

Drivers	IPPO SR	IPPO km	MAPPO SR	MAPPO km
20	26.1%	66.9	25.9%	90.7
23	29.4%	70.2	28.7%	102.3
26	32.6%	81.2	31.9%	121.0
29	35.4%	83.4	35.6%	137.6
32	38.4%	85.3	38.1%	155.7
35	40.5%	92.3	41.4%	174.0

Service rate rises ∼2.9 pp per 3 drivers (IPPO) / ∼3.1 pp (MAPPO); empty km rises ∼5.1 km per 3 drivers (IPPO) / ∼16.6 km (MAPPO). More drivers unambiguously buy more service at a proportional deadhead cost; there is no single optimum without a service-versus-empty-km tradeoff call. IPPO and MAPPO track each other closely on SR at every fleet size, but MAPPO spends substantially more deadhead (92 vs 174 at 35 drivers).

6.2 Reward-weight check

Measured on Bandra over the five IPPO-200 and five MAPPO-200 training seeds (3 episodes per seed, seeds 1000–1002, deterministic argmax; sample std across seeds):

	IPPO-200	MAPPO-200
relocations / episode	55.9 ± 10.7	58.3 ± 5.1
mean relocation leg (km)	1.40 ± 0.08	1.39 ± 0.09
pickup deadhead (km/completion)	0.85 ± 0.14	0.83 ± 0.07
P(credit fires | relocation)	0.683 ± 0.020	0.683 ± 0.020

With $\beta = 0.1$ and the measured 1.40 km leg, the per-leg break-even is $0.1 \times 1.40 = 0.140$. The bonus is share $\times\, \gamma = 0.65 \times 5.0 = 3.25$, so the literal margin is $3.25/0.140 = 23.2\times$; conditioning on the measured 0.683 fire rate gives

$$3.25 \times 0.683/0.140 = 15.9 \times .$$

Charging the whole job (leg plus 0.85 km pickup deadhead, 2.25 km) gives $0.1 \times 2.25 = 0.225$ and a conditional margin of $9.9\times$. MAPPO's measurements give $16.0\times$ (leg) and $10.0\times$ (whole job). Relocation is therefore strongly net-positive on Bandra; churn is not explained by a negative expected value.

The 2.95 km leg-plus-pickup figure is an NYC16 measurement (367 legs) and must not be used for Bandra. On NYC16 ($\beta = 0.06$, $\gamma = 1.0$) the per-leg break-even is $0.06 \times 2.95 = 0.177$, so the 0.65 share clears it $3.67\times$ literally and $2.09\times$ conditionally ($0.65 \times 0.568/0.177$).

6.3 Reward arithmetic check

Bandra uses $\alpha = 0.1$, $\beta = 0.1$, $\gamma = 5.0$, cancel $= 0.5$, with the relocation bonus paid on top of γ. NYC16 uses $\alpha = 0.2$, $\beta = 0.06$, $\gamma = 1.0$, with the bonus carved out of γ. Zone rewards sum to the team objective exactly, and the conditional form share $\times\, \gamma \times P(\text{dispatch}) > \beta \times \text{km}$ is asserted with the measured 0.568 NYC16 dispatch rate as an explicit input.

7 RL vs. Baselines

7.1 Protocol

Bandra, 25 zones, 30 drivers, 12 evaluation episodes each, deterministic argmax. Baseline evaluation uses the paired base-seed stream 9000–9011 (identical demand windows across Greedy and LP, 12 episodes each). The RL arms use `final_validation.json` from 5 seeds each (eval seed 9000, 12 episodes per seed, the same demand window across all 5 seeds). The two protocols are not paired: the baseline comparison uses 12 distinct demand windows, while the RL evaluation uses one repeated window across 5 training seeds, and the standard-deviation denominators differ (5 seeds vs 12 episodes).

7.2 Artifact

`results/bandra_baselines_D95.json` and `.csv`, produced by `scripts/eval_bandra_baselines.py` (12 episodes, seeds 9000–9011). Per-episode KPIs for Greedy and LP are on disk and reproducible; the RL arms are the `runs/*-bandra-*/final_validation.json` files.

7.3 Measured results

Method	SR	Empty km	Wait
Greedy	34.0% ± 2.7	20.7 ± 1.5	1.08 ± 0.17 min
LP	39.7% ± 2.2	98.4 ± 9.8	2.15 ± 0.17 min
IPPO-200	34.1% ± 1.0	170.6 ± 14.4	2.2 ± 0.16 min
MAPPO-200	35.1% ± 0.3	167.8 ± 6.8	2.2 ± 0.12 min

7.4 Headline

RL is within noise of Greedy on service rate at $\sim 8\times$ greedy's deadhead cost. LP dominates RL: 39.7% vs ∼34% SR at $0.58\times$ RL deadhead. On SR per 100 empty km the ranking is Greedy 164.2, LP 40.3, IPPO 20.0, MAPPO 20.9: Greedy tops this efficiency metric because it barely relocates, while LP buys its higher SR with $\sim 4.8\times$ Greedy's deadhead. The metric therefore supports "LP is less deadhead-hungry than RL", not "LP is the most efficient policy".

7.5 Conclusion

RL buys no service over greedy while spending $\sim 8\times$ greedy's deadhead; this is not an equal-budget comparison, so it does not establish that RL would lose at matched deadhead. LP dominates RL on the tradeoff frontier (higher SR at less than 60% of the deadhead). RQ1/RQ2 therefore cannot claim "RL beats baselines"; the supported claim is the tradeoff frontier above.

7.6 Standard deviations

All standard deviations in this report are sample std (ddof = 1) across seeds or episodes. The RL-arm SR stds are ±1.0 (IPPO-200) and ±0.3 (MAPPO-200); the corresponding population stds (ddof = 0) are ±0.9 and ±0.3. The table above uses sample std throughout; the conclusion is unchanged.

8 Discussion and Limitations

8.1 RL buys no service over greedy at roughly $8\times$ the deadhead

The central finding is that the learned policies match greedy on service rate while spending roughly $8\times$ as much deadhead distance on the five-seed final-validation protocol (170.6 vs 20.7 km). Section 4 documents that this multiplier is evaluation-dependent: the same IPPO-200-s0 checkpoint is 95.1 km on the pilot windows and 83.4 km in the 29-driver sweep. The LP baseline achieves higher service rate than RL while spending *less* deadhead than RL (98.4 km vs ∼170 km), so it dominates RL on the tradeoff frontier. On SR per 100 empty km the ranking is Greedy 164.2, LP 40.3, IPPO 20.0, MAPPO 20.9: Greedy tops that efficiency metric because it barely relocates, so the metric does not make LP the most efficient policy. The "RL beats baselines" claim is not supported; the supported claim is that LP dominates RL and RL buys nothing over greedy.

8.2 Candidate explanations for why RL does not win

1. **Observability gap.** The LP forecasts with an EMA over all past turns plus a 5-minute lookahead; the actor sees only 2 minutes of demand history (4 turns at $dt = 0.5$). The LP has an order-of-magnitude advantage in temporal horizon.

2. **Reward shape does not carry over.** On Bandra (5 training seeds × 3 episodes) the measured shares of reward movement are: served credit 59.0% ± 3.0 / 59.0% ± 1.1 (IPPO/MAPPO), relocation bonus 21.2% ± 4.4 / 22.3% ± 2.8, waiting 10.4% ± 1.5 / 9.9% ± 1.3, empty-km cost 2.2% ± 0.4 / 2.3% ± 0.4, cancellation 6.5% ± 1.1 / 6.3% ± 1.1. Served credit dominates; waiting is ∼0.17× the served signal, not 15×. The NYC16 90/6/2 shares do not apply because $\gamma = 5$ and the Bandra configuration pays the relocation bonus on top of γ rather than carving it out, so the reward-shape explanation is not evidence about the Bandra policies.
3. **Expected-value invariant was broken.** The relocator paid for the whole job it caused without earning proportionally, and credit was conditional on dispatch (0.68 on Bandra, 0.57 on NYC16). This was fixed by raising the relocator share to 0.65.
4. **Targeting is weak on Bandra.** Argmax picks the best legal target on 5.1% ± 6.9 of relocations for IPPO-200 and 8.8% ± 6.6 for MAPPO-200, against chance 5.4% ± 1.0 and 5.2% ± 0.7 over 20.7 and 21.1 legal targets per move. The chosen target holds 23% / 30% of the unmet demand available at the best legal target (demand ratio 0.23 / 0.30), and only 36% / 40% of relocations target a zone with any unmet demand. This is a Bandra measurement: IPPO and MAPPO are both consistent with chance; the NYC16 51% figure does not transfer.

8.3 Limitations

- Bandra RL checkpoints cannot load on NYC16 (different observation dimension and zone count). No NYC16 RL results are reported. The two tables are not directly comparable. On Bandra, Greedy vs LP uses paired demand windows (base seeds 9000–9011), but the RL arms are evaluated separately, so the RL-versus-baseline comparison is not paired.
- Final-iteration validation versus best-checkpoint selection: optimistic selection would inflate RL by ∼1 pp.
- Reward-share and targeting numbers are 5 training seeds × 3 episodes each, deterministic argmax.

8.4 What would move the needle

1. **Longer demand history.** The actor sees 6.7% of the episode (4 of 60 turns); the LP has an EMA over all turns plus a 5-minute lookahead.
2. **Credit for good targeting.** Scale relocator credit by the deficit relieved at the destination, not a flat share.
3. **Lower α or cap the waiting term.** On Bandra waiting is 10.4% / 9.9% of reward movement, ∼0.17× the served signal, so this is a much weaker lever than the NYC16 numbers suggested.
4. **Temperature on destination logits.** The entropy coefficient is a dead lever (93–94% max entropy in the relocate block); a temperature on the destination logits would sharpen the argmax without changing it.

8.5 Figures

All three degradation plots are regenerated from `results/shift_eval/summary.json`. Each figure has two panels: left, Bandra RL arms (5 seeds); right, NYC16 baselines. The environments are not comparable, so lines must not be compared across panels. The Bandra arms hold ∼35–38% SR across S0, S1, S2a–S2e, S3, S5, S6, S7, with only S4 (cold-zone concentration) collapsing to ∼21%; S6 equals S0 on Bandra because the test-split switch is a no-op there. The NYC16 baselines sit at different levels (Greedy ∼23–50%, LP ∼47–80%) and are shown only for completeness.

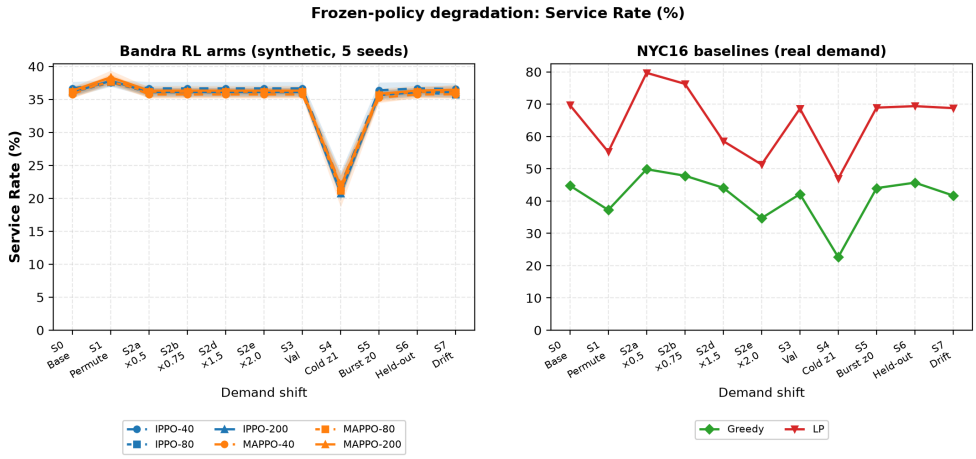


Figure 1: *Frozen-policy service-rate degradation. Left: Bandra RL arms (5 seeds). Right: NYC16 baselines. Panels are separate environments and are not directly comparable. Bands are ±1 sample std.*

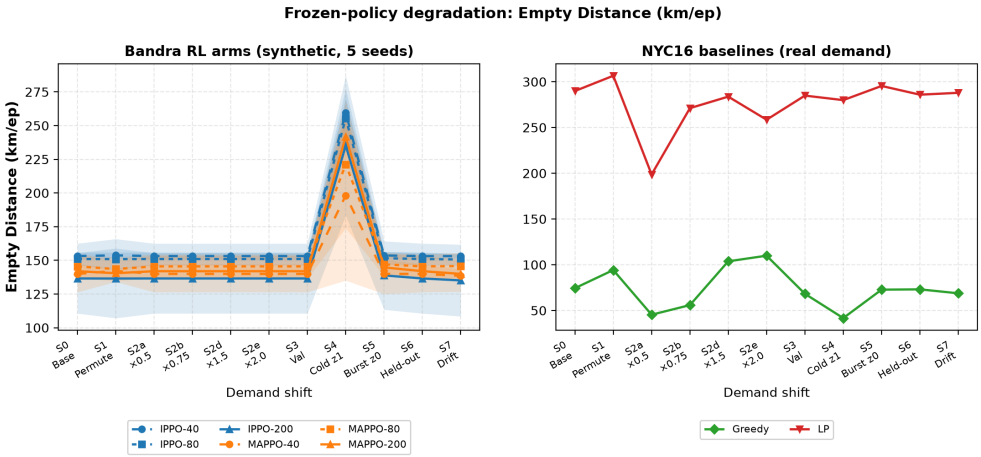


Figure 2: *Empty-distance degradation. Left: Bandra RL arms (5 seeds). Right: NYC16 baselines. Panels are separate environments and are not directly comparable. Bands are ±1 sample std.*

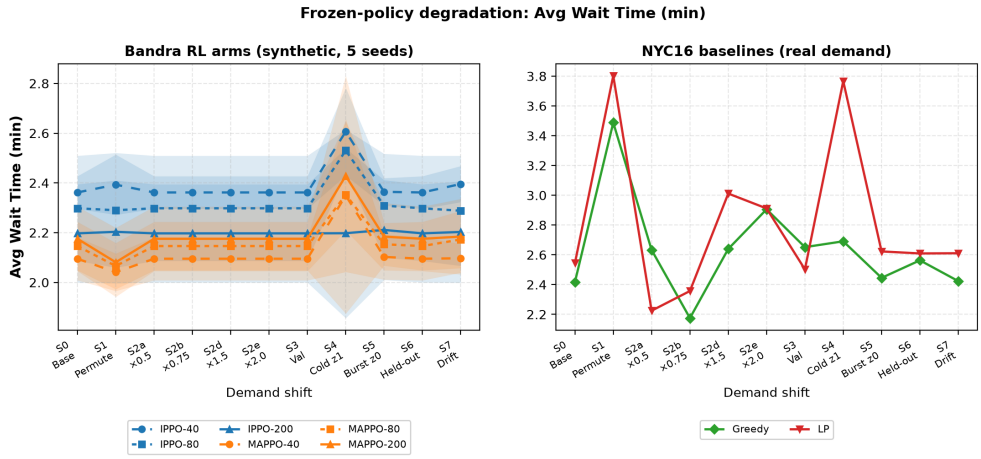


Figure 3: *Average wait-time degradation. Left: Bandra RL arms (5 seeds). Right: NYC16 baselines. Panels are separate environments and are not directly comparable. Bands are ±1 sample std.*

Reproducibility

All figures and tables are generated from artifacts in the repository:

- `runs/*-bandra-*/final_validation.json` – 30 training cells
- `results/bandra_baselines_D95.json` – Greedy/LP baselines
- `results/bandra_fleet_D93/` – fleet-size sweep
- `results/bandra_argmax_pilot_D91/` – argmax-vs-sampled pilot
- `results/shift_eval/summary.json` – S0–S7 shift evaluation
- `results/bandra_terms.json` – Bandra reward-share, targeting and relocation measurements, 5 seeds × 3 episodes (per-seed files alongside; `scripts/diag_bandra_terms.py`)
- `docs/figures/` – regenerated degradation figures

Artifact filenames retain internal experiment IDs. Code and artifacts: https://github.com/sharifmdathar/marl-fleet-rebalancing.

References

[1] X. Guo, N. S. Caros, and J. Zhao. Robust matching-integrated vehicle rebalancing in ride-hailing system with uncertain demand. *Transportation Research Part B: Methodological*, 150, 2021.

[2] J. K. Terry et al. PettingZoo: Gym for multi-agent reinforcement learning. *NeurIPS*, 2021. arXiv:2009.14471.